

Owner override and guided-practice contract

LEARNER CONDITIONING

Student knowledge percentage was not supplied. This review fixture uses an explicit owner override; no learner percentage is fabricated. The override changes support only, not Chemistry truth or frozen question identity.

CORE2A JOB

Declarative deliberate practice: attempt first, then technical clue, setup, quick check, full solution, error classification and retry. Questions remain frozen.

Episode 1 | WORKED | RQ-REDOX-01

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

ATTEMPT

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, prove which species is oxidised and which is reduced using oxidation-state changes.

WORKED REPAIR

Zn: 0 -> +2, so Zn is oxidised. Cu in Cu^{2+} : +2 -> 0, so Cu^{2+} is reduced. Technical reason: compare the same element and use increase/decrease, not reagent reputation.

Episode 1 | Error classification and retry

IF WRONG

Species mismatch -> SPECIES_IDENTITY_ERROR. Reversed increase/decrease -> CHARGE_OR_OXIDATION_STATE_ERROR. Correct label without proof -> CONCEPT_MODEL_ERROR.

RETRY

Cover the worked repair and reconstruct both lanes again from the reaction only.

Episode 2 | COMPLETION | RQ-REDOX-02

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

COMPLETE

$\text{Zn} \rightarrow \text{Zn}^{2+} + \underline{\hspace{1cm}}$; $\text{Cu}^{2+} + \underline{\hspace{1cm}} \rightarrow \text{Cu}$. Then identify both agents and justify each role from the self-change.

FULL SOLUTION

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$. $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$. Zn is the reducing agent; Cu^{2+} is the oxidising agent.

Episode 2 | Error classification and near retry

IF WRONG

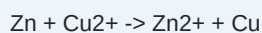
Wrong electron side -> SYMBOLIC_EXECUTION_ERROR. Wrong count -> CONSERVATION_ERROR. Reversed agent role -> CONCEPT_MODEL_ERROR.

RETRY

Write a four-column ledger: species | state change | electron change | role. Fill role last.

Episode 3 | INDEPENDENT | RQ-REDOX-03

GOVERNED REACTION



ATTEMPT WITHOUT HELP

A learner says: 'Zn is the reducing agent because Zn gets reduced.' Diagnose the error and repair the reasoning.

QUICK CHECK

If your first sentence says Zn is reduced, reopen only the oxidation-state lane before reading the full repair.

Episode 3 | Complete worked repair

WORKED REPAIR

The claim reverses Zn's self-change. Zn changes 0 -> +2, so Zn is oxidised; because its electron loss reduces Cu^{2+} , Zn is the reducing agent.

WHY THE ERROR PERSISTS

The label 'reducing agent' describes what the species causes elsewhere, not the species' own Redox process. Separate SELF from OTHER.

Exit check | no mastery claim

INDEPENDENT CHECK

Without looking back, write the full proof and then explain why 'reducing agent' cannot be translated as 'gets reduced'.

NON-CLAIM

Completing this printed episode does not prove mastery or retention. It is an opportunity to demonstrate the reasoning; learner evidence must be recorded elsewhere.